

National University of Singapore
Department of Computing

BT2102 Data Management and Visualisation

**Tutorial 6:
Data Storytelling**

BACKGROUND

Assuming you are a researcher in a government agency given the datasets in “**appointment.xlsx**” which contains information about medical appointments from public healthcare institutions.

Several characteristics are represented in the columns and these characteristics about a certain patient and his/her appointment are included in each row.

Focus is on the question of whether patients show up for their appointments. No-shows, or patients who missed their scheduled appointments are common and costly to healthcare institutions. Medical resources are not efficiently utilized, and delays are caused by others in need of medical resources, which may increase the possibility for adverse health outcomes. A No-show data analysis can help healthcare institutions understand what attributes are associated with no-show situation and predict whether a patient will break the appointment to prevent no-show in different ways (e.g., reminder phone calls, double-book an appointment slot, schedule at a different time for certain patient).

Dataset Details

Here is the description of dataset features:

Column	Description
PatientId	Unique identification of a patient
AppointmentID	Unique identification of each appointment. A patient may have multiple appointments.
Gender	M - male F - female
ScheduleDay	The time when patient called to schedule an appointment.
AppointmentDay	The day of the actual appointment, when patient should visit the doctor.
Age	How old is the patient?
Neighbourhood	In which city the appointment takes place.
FamilyAllowance	The Government offers financial aid to very poor family. 0 - not receive financial aid 1 - receive financial aid
Hypertension	0 - not suffer from hypertension 1 - suffer from hypertension
Diabetes	0 - not suffer from diabetes 1 - suffer from diabetes
Alcoholism	0 - not suffer from alcoholism 1 - suffer from alcoholism
Handicap	0 - not handicapped 1 – handicapped
SMS-received	Healthcare institutions will send one or more messages to patients to inform them of their appointment. 1 - at least one message received by the patient 0 - no message received by the patient
No-show	Yes - patient broke the appointment No - patient kept the appointment

WHAT ARE YOU REQUIRED TO DO?

Consider the business-critical level-1 question to be: **“What are the factors contributing to the number of No-show?”**

Given this level-1 question, you may consider some of these level-2 and possible level-3 questions as part of your analysis to answering the level-1 question:

1. Is living in certain city or neighborhood a factor contributing to no-show?
 - Level-3 question: What are the neighborhoods with high no-show?
 - Level-3 question: How do neighborhoods receiving financial aid from government perform in the number of no-show?
2. Is there a relationship between Age and no-show?
3. Do patients with medical conditions contribute to the number of no-show?
 - Level-3 question: Which neighborhood has high concentration of patients with hypertension contributing to no-show?
 - Level-3 question: Which neighborhood has high concentration of patients with diabetes contributing to no-show?
4. Do people with handicaps contribute to the number of no-show?
5. Does long wait time contribute to the increase in number of no-show?
6. Does sending sms notification as a reminder of appointment help in improving no-show numbers?
 - Level-3 question: What is the insight between those who received sms and turn-up rate and those who did not receive sms and turn-up rate?
 - Level-3 question: Does Age group matter in the no-show numbers even when sms is received?

Prepare your data visualization for the above level-2 or level-3 questions and produce data insights and recommendations as part of the storytelling.

You will be asked to show your data visualisations and explain your data insights including recommendations where applicable during the tutorial session.